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SUBJECT CODE NO: - RNP_8008
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A & B) (NEP) Sem - I and Sem - II
Examination November / December 2025 (To be held in February-2026)
Engineering Physics

[Time: 3 Hours]**[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no. 1 is compulsory
- 3) Solve any four questions from Q. nos 2, 3, 4, 5, 6 and 7

Section-A

Q1 Solve the following questions (any 10) 20

a) Newton's rings are formed due to:

- A) Diffraction B) Polarization C) Interference D) Reflection**

b) Type-II superconductors are also known as:

- A) Hard superconductors B) Soft superconductors C) Perfect conductors D) Magnetic insulators**

c) Michelson interferometer works on the principle of:

- A) Diffraction B) Interference C) Refraction D) Photoelectric effect**

d) Ultrasonic waves have frequencies.....

e) Bragg's law is given by.....

f) What is interference of light?

g) Define De-Broglie wavelength

h) Write any two applications of X-ray diffraction (XRD).

i) What is quantum tunneling

j) Define single-mode optical fiber

k) A He-Ne laser emits light of frequency 4.74×10^{14} Hz. Calculate the wavelength.

l) The energy level of electrons at absolute zero is called _____.

Section-B

Q2 A. Explain the formation of Newton's rings and derive an expression for the radius of bright and dark rings. 05

B. Write a short notes on: i) Electromagnetic waves ii) Electromagnetic spectrum 05

Q3 A. Explain the production and detection of X-rays. Discuss continuous and characteristic spectra. 05

B. Explain Davisson-Germer experiment 05

- Q4 A. Explain superconductivity, Meissner effect and discuss Type-II superconductors 05
- B. Explain structure of optical fiber with suitable diagram, refractive index of core 1.48 and that of cladding is 1.47 in an optical fiber calculate numerical aperture 05
- Q5 A. Describe the construction, working, and applications of the Michelson interferometer. 05
- B. Explain the Photoelectric effect with diagram. 05
- Q6 A. Derive Bragg's law and explain its applications in X-ray diffraction and lattice parameter determination 05
- B. The spacing between principal planes of crystal is $2.82 \text{ \AA}$. The first order Bragg reflection occurs at an angle of 9° . What is the wavelength of X-rays? 05
- Q7 A. The wavelength of the X-rays is 0.07 nm which is diffracted by a plane of salt with 0.28 nm . As the lattice constant. Determine the glancing angle for the second order diffraction. Assume the value of the salt flame to be 110 and the given salt is Rock Salt 05
- B. Explain the ruby Laser? 05